

Impetus — AI Design Agent

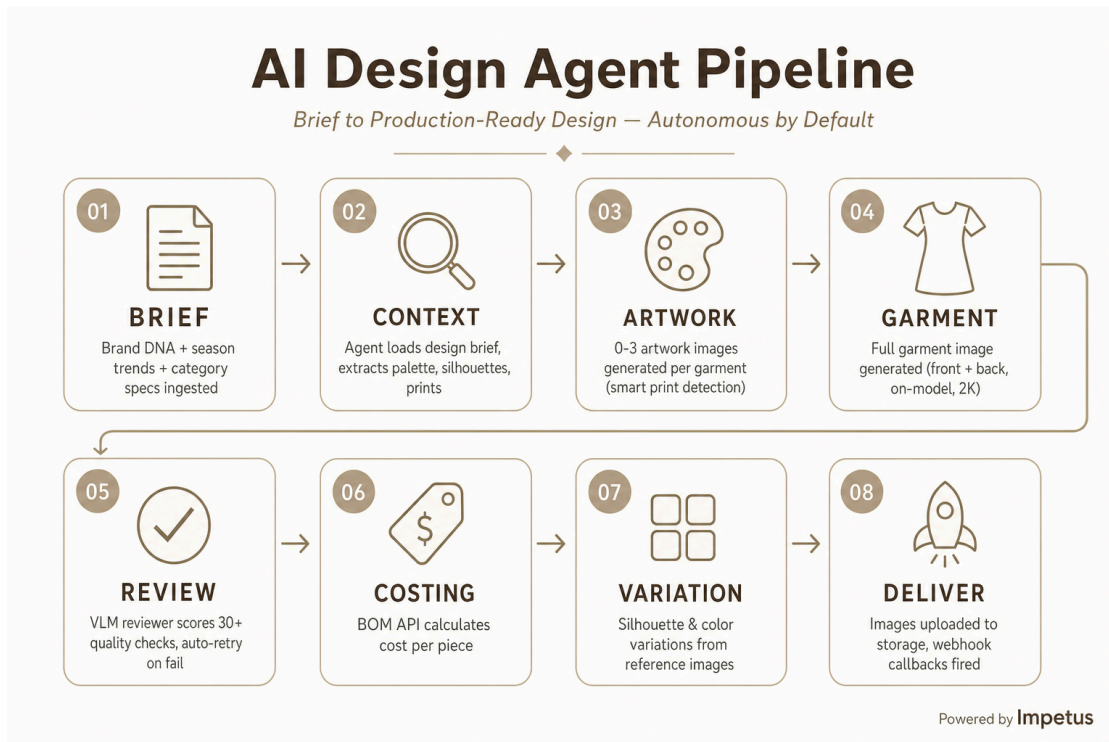
Agentic AI for Production-Ready Garment Design

1) What it does

Impetus AI Design Agent is an autonomous, multi-agent system that generates production-ready garment designs from trend briefs. It takes brand DNA, season trends, and category specifications as input — and outputs complete garment images (front + back, on-model, 2K resolution) with artworks, swatch panels, and BOM costing. The pipeline runs end-to-end without human-in-the-loop; human review happens only at selection/approval.

Three modes: Hybrid (artwork + garment — fastest), Standard (full 4-image design package), and Variation (structural/color variations from reference images for range building).

2) The agentic design flow



- **Brief Ingestion** — Brand DNA, season direction, category specs, and moodboard palette ingested from the trend engine.
- **Context Agent** — Loads the design brief, extracts brand palette, silhouette direction, print techniques, and fabric preferences.
- **Artwork Generation** — 0-3 artwork images per garment with smart print detection. Solid garments skip; printed garments get placement or AOP artworks with VLM quality review.
- **Garment Image** — Full garment image (front + back, on-model). Zone-aware artwork placement, gender-correct model, age-appropriate styling.
- **Quality Review** — VLM reviewer scores 30+ checks (color accuracy, print fidelity, silhouette, category). Auto-retries on failure.
- **BOM Costing** — Cost per piece calculated via costing API (fabric, trims, SMV, overheads).
- **Variation Engine** — Generates N silhouette and color variations from reference images using moodboard palette. Used for range building and style explosions.

- **Delivery** — Images uploaded to cloud storage via presigned URLs. Per-style webhook callbacks fired in real time.

3) Category coverage — F&L operating model (43 pods)

Mapped against the 43 category pods defined in the F&L operating model. Coverage status and enhancement plan for each pod below.

Men's Apparel (Pods 1-9)

#	Category Pod	Covered	Enhancement Plan
1	Men Western: Denim Bottoms	Yes	—
2	Men Western: Non-Denim Bottoms	Yes	—
3	Men Western: Casual T-shirts	Yes	—
4	Men Western: Casual Shirts	Yes	—
5	Men Western: Formal Shirts	Yes	—
6	Men Western: Formal Trouser & Suits	Yes	—
7	Men Indian: Festive Wear	Yes	Improving craft detail accuracy for sherwanis
8	Men: Sports / Athleisure	Partial	Good at basic designs. Improving for dynamic, complicated choice of fabrics (by May 28)
9	Men: External Brands	N/A	N/A

Women's Apparel (Pods 10-18)

#	Category Pod	Covered	Enhancement Plan
10	Women: Western Tops	Yes	—
11	Women: Western Dresses, Jackets, Blazers	Yes	Improving complex drape constructions (by May 25)
12	Women: Western Denim Bottoms	Yes	—
13	Women: Western Non-Denim Bottoms	Yes	—
14	Women Indian: Kurta/SKD/Bottoms	Yes	—
15	Women Indian: Saree/Lehenga	Partial	Saree draping pipeline in development (by May 30)
16	Women Indian/Fusion: Festive SKD	Yes	—
17	Women: Sports / Athleisure	Partial	Good at basic designs. Improving for dynamic, complicated choice of fabrics (by May 28)
18	Women: External Brands	N/A	3P marketplace brands

Kids Apparel (Pods 19-21)

#	Category Pod	Covered	Enhancement Plan
19	Kids: Infant	Yes	—
20	Kids: Boys	Yes	—
21	Kids: Girls	Yes	—

Footwear & Accessories (Pods 22-28, 34)

#	Category Pod	Covered	Enhancement Plan
22-25	Footwear (Sports, Men, Women, External)	No	3D footwear pipeline — scoping phase (by May 30)

26	Innerwear: Men & Women	Partial	Basic styles covered. Expanding to structured innerwear (by May 25)
27	Accessories: Gold & Jewellery	No	Requires accessories pipeline
28	Accessories: Others (Bags, Belts, Wallets)	No	Requires accessories pipeline
34	Beauty	No	Out of scope — separate division

Quick Fashion (Pods 29-33)

#	Category Pod	Covered
29	Quick Fashion Women	Yes
30	Quick Fashion Men	Yes
31	Quick Fashion Kids	Yes
32	Quick Fashion Footwear	No — depends on footwear pipeline
33	Quick Fashion Accessories	No — depends on accessories pipeline

Premium / BTL & Luxury (Pods 35-43)

#	Category Pod	Covered	Enhancement Plan
35	Premium Men	Yes — Variation Generator	Buyers upload references; agent generates on-brand variations
36	Premium Women	Yes — Variation Generator	Reference-based style explosions maintaining brand DNA
37	Premium Kids	Yes — Variation Generator	Age-appropriate variations from reference images
38	L&BTL Men	Yes — Variation Generator	Range building with Pantone-accurate colors
39	L&BTL Women	Yes — Variation Generator	Range building with Pantone-accurate colors
40	Footwear & Accessories	No	Requires separate pipeline (target: June)
41	Eyewear	No	Eyewear pipeline not yet scoped (target: June)
42	Indian Designer Wear	Partial	Lehengas, festive SKD covered. High-craft couture extending (by May 30)
43	Premium Home & Living	No	Home textiles pipeline not yet scoped

4) Execution track record

Trend Business — 718 Designs, Under 2 Days

Executed a live design explosion for the Trend business across 10 brands — DNMX (Men & Women), Netplay, Team Spirit (Men & Women), Rio, Avaasa Sets, Avaasa Mix n Match, Frendz Girls, and INF Frendz. 718 designs were generated in under 2 days with agents working autonomously across jeans, cargos, shirts, t-shirts, co-ord sets, kurta sets, kurtis, dresses, dungarees, jackets, hoodies, polos, and track pants.

Early signal: ~60% average selection rate across categories. Indian-wear (Avaasa) and Denims (DNMX) performed strongest. The design team used these outputs as the foundation for building the full range architecture, with V2 iterations driven by finalized fabric lists.

Impetus x Superdry India — 20 Designs Live on Retail

In February, our team visited the Superdry India Delhi office to kickstart what became the SDX Spring/Summer '26 capsule. The project formally began in April, and in under 10 days we had over 400 designs ready. From those, 171 pieces were selected for the SDX SS '26 capsule, and 20 designs have been launched by Superdry India — now live on retail as part of their SS'26 drop on Ajio.

From bold Sui Riyu water-dragon graphics to hyper-charged Mecha Street gaming visuals and dopamine-fuelled Lucid Rush prints, this capsule showcases the power of Impetus's design engine, SDX-focused trend research, and tight cross-team collaboration. The entire set of 400+ designs was created in under 10 days.

5) Eval approach

Automated VLM-based evaluation scores every generated garment across 6 quality dimensions (0-100 scale), evaluated against the actual design brief and generation prompt. Gives per-garment, per-category, and per-brand quality tracking.

- **Overall score:** 83/100 average across 2,076 garments, 29 brands
- **Production-ready:** 92% of categories score 75+
- **Scoring dimensions:** Category accuracy (94), Prompt adherence (87), Brand identity (84), Trend alignment (84), Commercial viability (83), Brief adherence (79)
- **Key insight:** Category accuracy is strongest (94) — pipeline consistently generates the correct garment type. Brief adherence (79) is the active improvement area.

6) What's next

- **Variation engine at scale** — Reference-based range building now live. Generates colorway and silhouette variations with Pantone-accurate moodboard colors.
- **Premium brands via variations** — Premium and L&BTL pods (35-39) served through the Variation Generator. Buyers upload existing brand references, and agents generate on-brand variations.
- **Trend-to-Design closed loop** — Trend Agent outputs feed directly into the Design Agent. Fully autonomous brief-to-design pipeline.
- **Delta categories** — Non-apparel pods (footwear, accessories, eyewear, home) require category-specific pipelines. Scoping underway for June.
- **Eval-driven improvement** — Continuous automated evaluation on every generation. Weakest dimension (brief adherence at 79) is the active target.

Improvement Timeline (May 21 - May 30)

- **May 21-25:** Complex drape constructions for dresses (pod 11), structured innerwear expansion (pod 26)
- **May 25-28:** Dynamic athleisure fabric handling for sports/athleisure pods (pods 8, 17)
- **May 28-30:** Saree draping pipeline (pod 15), high-craft Indian designer wear (pod 42), sherwani craft details (pod 7)